



Power BI Project Report (week 2) –

Superstore Sales Analysis

1. Introduction

The objective of this project was to analyze sales data using Power BI. The tasks included importing data, cleaning it, creating relationships, building measures with DAX, and designing an interactive dashboard to visualize insights.

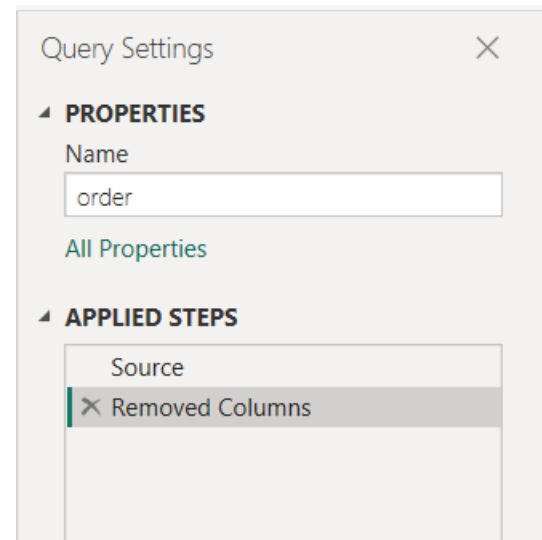
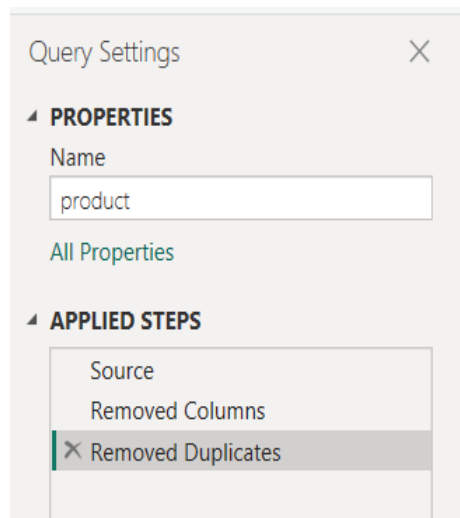
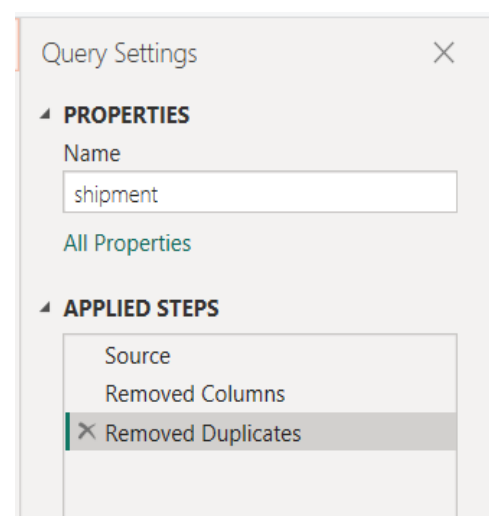
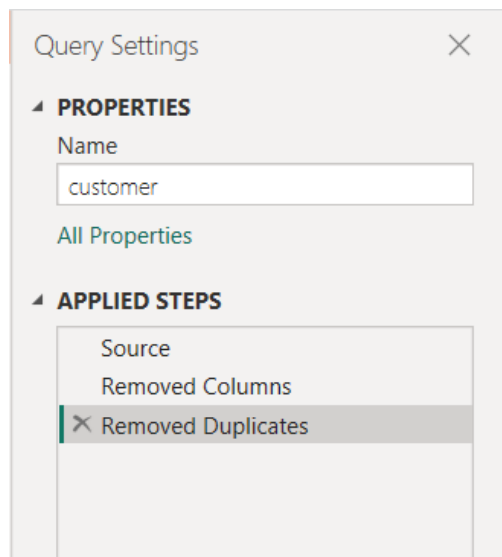
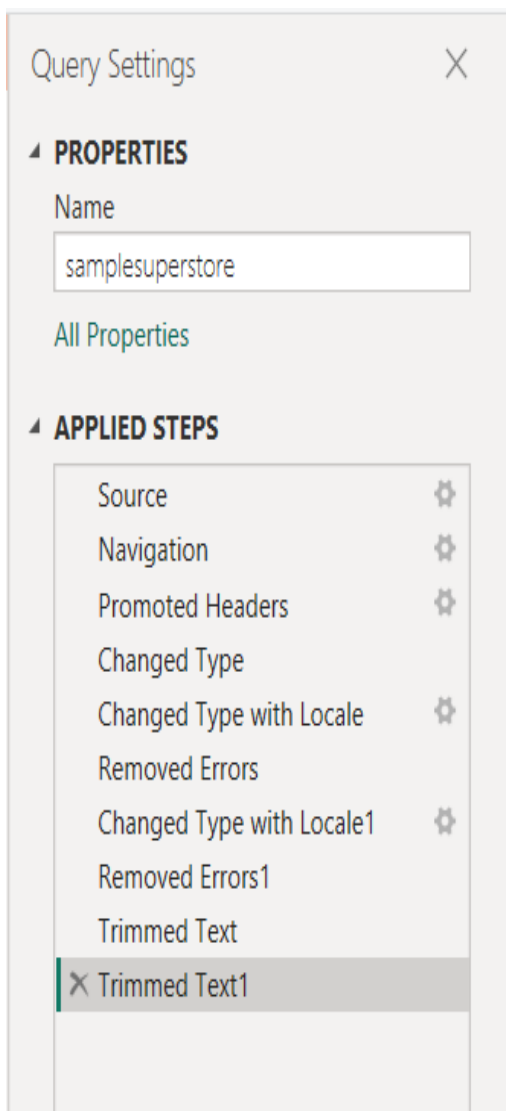
2. Data Import

- Imported the dataset (Orders.csv, Shipments.csv) into Power BI.
- Verified column names and data types.

samplesuperstore Category City Country/Region Customer ID Customer Name Σ Discount > Order Date Order ID Postal Code Product ID Product Name Σ Profit Σ Quantity Region Σ Row ID Σ Sales Segment > Ship Date Ship Mode State/Province Sub-Category	customer Customer ID Customer Name Region Segment > DataTable > Date MonthName MonthNumber Orders Card Profit Card Quantity Card Quarter Sales Card Year shipment Order ID Region > Ship Date Ship Mode	order Customer ID Σ Discount Order Date Order ID Product ID Σ Profit Σ Quantity Σ Sales > Ship Date Total Orders Total Profit Total Quantity Total Sales > product Category Product ID Product Name Sub-Category
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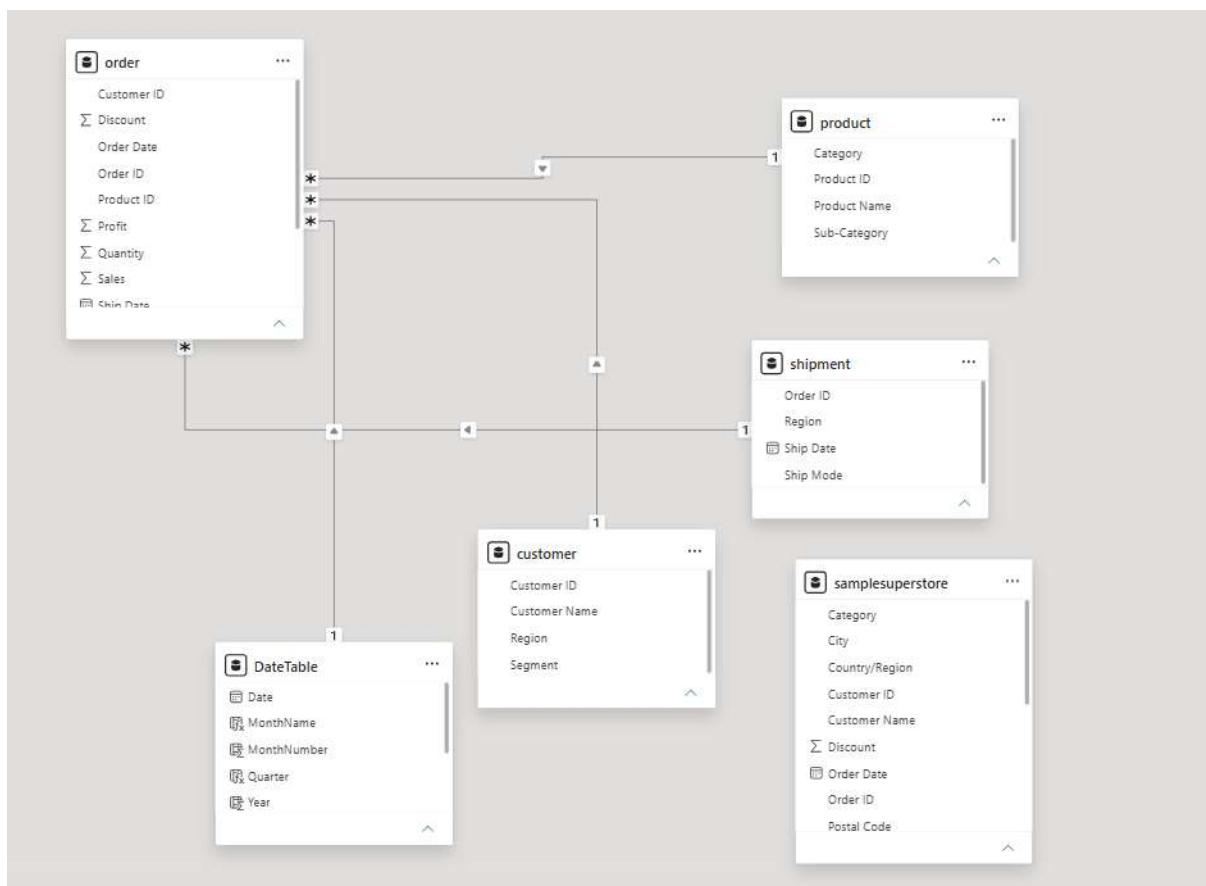
3. Data Cleaning (Power Query)

- Renamed columns for clarity.
- Changed data types (e.g., Order Date → Date).
- Removed duplicates.
- Extracted Year, Month, and Quarter from Order Date.



4. Data Modeling

- Created a **DateTable** using **CALENDAR** function.
- Marked DateTable as a Date Table.
- Established relationships:
 - **Orders[Order Date] → DateTable[Date]**
 - **Orders[Ship Mode] → Shipments[Ship Mode]**



5. Creating Measures (DAX)

Defined key measures for analysis:

DAX

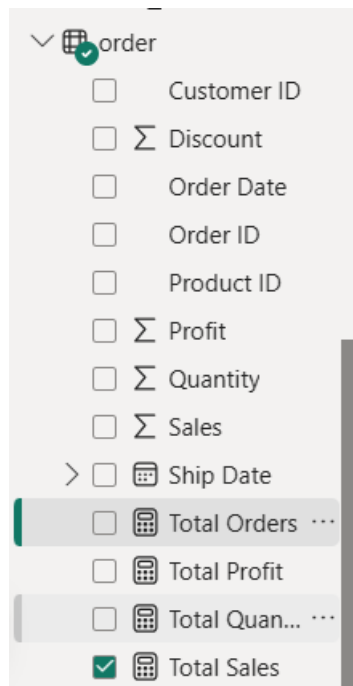
Total Sales = SUM('order'[Sales])

Total Profit = SUM('order'[Profit])

Total Orders = COUNTROWS('order')

Total Quantity = SUM('order'[Quantity])

Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)



6. Dashboard Design

- Placed KPI cards at the top:
 - 💰 Total Sales
 - 📦 Total Quantity
 - 📈 Total Profit
 - 📄 Total Orders
- Added visuals:
 - Line chart: Sales by Month

- Bar chart: Profit by Category
- Pie chart: Sales by Segment
- Added slicers for Date, Region, and Segment.



7. Analysis & Insights

- **Sales Trend:** Sales peaked in February–March, then declined toward December.
- **Category Profitability:** Technology generated the highest profit, followed by Office Supplies and Furniture.
- **Customer Segments:** Consumer segment contributed nearly half of total sales, Corporate ~33%, Home Office ~20%.

